

Homework Problems. *Solutions should be written/typed in numerical order and clearly labeled. Unsolved problems should be included but left blank.*

1. Let $(\mathbb{Q}, +)$ be the group of rational numbers under addition.
 - a. Is $\mathbb{Z}$ a subgroup of $(\mathbb{Q}, +)$?
 - b. Is $\mathbb{Z}^+ = \{0, 1, 2, 3, \dots\}$ a subgroup of $(\mathbb{Q}, +)$?
 2. Determine if the following are subgroups.
 - a. Is $\mathbb{Q}$ a subgroup of the group, $(\mathbb{C}, +)$, the complex numbers under addition?
 - b. Is $\mathbb{Z} - \{0\}$ a subgroup of the group, $(\mathbb{Q} - \{0\}, \cdot)$, the non-zero rational numbers under multiplication?
 3. Prove that the intersection of two subgroups H and K of a group G is a subgroup.
 4. Let $H_1, H_2, \dots$ be subgroups of G . Prove that $H_1 \cap H_2 \cap \dots$ is a subgroup of G .
 5. Recall that $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ is a group under modular addition.
 - a. Show that $H = \{0, 2, 4\}$ and $K = \{0, 3\}$ are subgroups of $\mathbb{Z}_6$.
 - b. Show that $H \cup K$ is *not* a subgroup of $\mathbb{Z}_6$.
 - c. Is the union of two subgroups of G necessarily a subgroup G ?
 6. Let G be an abelian group and H be a subgroup of G . Prove that $S(H) = \{x | x \in G \text{ and } x \cdot x \in H\}$ is also a subgroup of G .
 7. Let G be a group. Prove that G has a subgroup with one element. This group is called the *trivial group*.
- Definition.** Let G be a group. If there is a single $g \in G$ such that every element in G is equal to g^n for some $n \in \mathbb{N}$ then G is a *cyclic* group and x is called the *generator* of G .
8. Prove that:
 - a. $\mathbb{Z}_n$ is a cyclic group under modular addition for any $n \in \mathbb{Z}$. (you can use the fact that it is a group).
 - b. $\mathbb{Z}$ is a cyclic group.
 9. Find all generators of
 - a. $\mathbb{Z}_6, \mathbb{Z}_8$, and $\mathbb{Z}_{10}$ (note: $\mathbb{Z}_n$ is often notated Z_n).

Definition. Let $X = \{1, 2, \dots, n\}$. The *symmetric group on X* , S_n , is the collection of all bijections $\sigma : X \rightarrow X$. That is, it is the set of all functions that rearrange the elements of X . (S_n is also called the *permutation group of the set X*).

10. Let S_3 be the symmetric group of permutations on three elements. Show that S_3 is:
 - a. Non-abelian.
 - b. Not cyclic.